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RESEARCH

Zero-shot performance of a general-purpose vision-language model for pediatric appendicitis diagnosis

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Abstract

Background General-purpose vision-language models can analyze medical images without task-specific training, but their value for pediatric abdominal ultrasound is unknown.

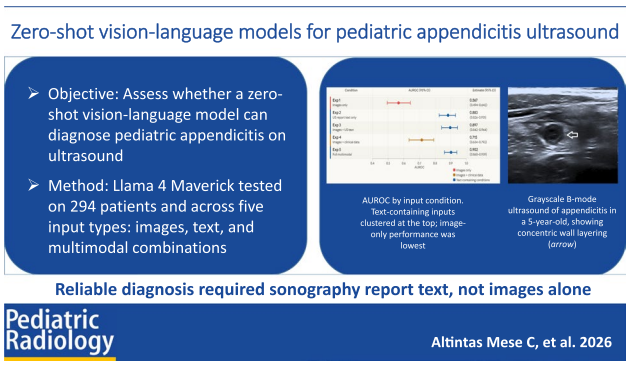
Objective To investigate the zero-shot performance of a general-purpose vision-language model to diagnose pediatric appendicitis using multimodal inputs, including images, report text, and clinical information.

Materials and methods In this retrospective study, diagnostic capabilities of Llama 4 Maverick were evaluated on the Regensburg Pediatric Appendicitis Dataset. Five experiments were conducted using the following inputs as well as input combinations: images only, ultrasound report text only, images and ultrasound text, images and clinical data, entire multimodal input. Reference standard of appendicitis was defined based on histopathology in patients who underwent surgical resection and on clinical follow-up in patients managed conservatively. Performance was assessed at the patient level using accuracy, sensitivity, specificity, positive predictive value, negative predictive value, and F1-score. Area under the receiver operating characteristic curve (AUROC) was evaluated secondarily as a measure of discrimination.

Results Of 782 patients, 294 met inclusion criteria; based on data availability, four experiments are conducted using 293 patients and in the fifth experiment, 284. Images-only experiment showed very low specificity (13.5%) and had the poorest discrimination (AUROC, 0.567). Ultrasound report text improved classification performance, with specificity increasing to 63.5% while maintaining high sensitivity; discrimination was also strong (AUROC, 0.883). Adding images to ultrasound text or combining all available inputs did not yield further meaningful improvement. Differences between images-only experiment and text-containing experiments were statistically significant ($P \leq 0.002$).

Conclusion Zero-shot visual interpretation of pediatric abdominal ultrasound by a general-purpose vision-language model is inadequate for safe appendicitis diagnosis. Performance is driven primarily by a structured sonographic imaging report, supporting a role for such models as text-based decision support tools rather than autonomous ultrasound image interpreters.

Graphical Abstract



Keywords Large language model · Multimodal artificial intelligence · Pediatric appendicitis · Ultrasound · Vision-language model

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Introduction

Acute appendicitis is the most common surgical emergency in childhood, with an estimated annual incidence of 70–94 cases per 100,000 children and a lifetime risk of approximately 7–8% [1, 2]. Timely diagnosis is critical because delayed appendectomy substantially increases the risk of perforation, which occurs in 20–35% of pediatric cases and is associated with prolonged hospital stay and increased morbidity [3]. Abdominal ultrasound is the recommended first-line imaging modality for suspected pediatric appendicitis in most international guidelines, owing to its absence of ionizing radiation, wide availability, and acceptable diagnostic performance in experienced hands [4, 5]. Established sonographic criteria include an outer diameter exceeding 6 mm, non-compressibility, target sign, appendicolith with acoustic shadow, periappendiceal fat stranding, and hyperemia on Doppler examination [6]. Structured clinical scoring systems, including the Alvarado score and the Pediatric Appendicitis Score, guide selective imaging, but their sensitivity and specificity remain insufficient to replace imaging in intermediate-risk patients [7, 8].

Artificial intelligence approaches to diagnostic imaging have expanded rapidly [9, 10], with supervised machine learning and deep learning models demonstrating strong diagnostic performance for appendicitis detection across multiple imaging modalities and patient populations, in some cases matching or exceeding the accuracy of clinical scoring systems [11, 12]. However, these models require large annotated training datasets and re-validation when imaging hardware or acquisition protocols change, limiting their practical deployability [12]. These limitations have increased interest in more flexible multimodal systems that can generalize across tasks without the need for dedicated retraining.

General-purpose vision-language models (VLMs) are large multimodal neural networks trained on image–text pairs that can perform medical image interpretation tasks without task-specific fine-tuning [13, 14]. This zero-shot capability eliminates the need for annotated training datasets. In the zero-shot paradigm, a model is applied to tasks it has not been explicitly trained or fine-tuned to perform, relying on knowledge acquired during broad pre-training on large-scale image–text data. Models including GPT-4V and GPT-4o have demonstrated zero-shot ability to describe radiological findings and classify pathology from medical images across CT, MRI, and plain radiography [15–19]. However, a consistent finding across this literature is that VLM performance is substantially stronger on text-based than image-based inputs, a pattern characterized as "textual proficiency and visual deficiency" in one systematic evaluation [20].

Performance benchmarking on ultrasound has been limited, and the modality presents particular challenges including acoustic artifacts, operator-dependent acquisition, and variable image quality not present in CT or MRI. No published study has evaluated a general-purpose VLM on pediatric abdominal ultrasound, and no study has systematically isolated the contribution of visual sonographic image content from structured text findings within a single controlled cohort. To address this gap, we conducted a modality ablation study evaluating the zero-shot diagnostic performance of Llama 4 Maverick, a publicly available multimodal large language model, for diagnosing pediatric appendicitis across five input conditions. We used the Regensburg Pediatric Appendicitis Dataset [21], which provides ultrasound images and structured tabular data for the same patients, enabling controlled within-cohort comparison of each information source. The primary objective was to isolate the independent diagnostic contribution of visual image content, structured sonographic imaging report, and clinical laboratory context to VLM-based appendicitis diagnosis.

Materials and methods

Study design

This retrospective study evaluated the diagnostic performance of a general-purpose VLM for pediatric appendicitis across multimodal input variations using abdominal ultrasound images, structured sonographic imaging report, and clinical laboratory data. The study was a secondary analysis of a publicly available, anonymized dataset [21]; model outputs did not influence clinical care at any stage. Additional institutional review board approval and informed consent were not required. Reporting followed the Reporting Checklist for Foundation and Large Language Models (REFINE) [22]. The checklist is provided on the Open Science Framework [23].

Dataset and study population

We used the Regensburg Pediatric Appendicitis Dataset [21] (version 1.0, published 23 February 2023, CC BY-NC 4.0), accessed on 15 March 2026. The dataset was collected at the University Children's Hospital Regensburg (KUNO) from children presenting with suspected appendicitis and includes pediatric abdominal B-mode ultrasound images together with tabular clinical data. For most patients, 1–15 ultrasound images are available, consisting of the right lower abdominal quadrant, appendix, intestines, lymph nodes, reproductive organs, and regions of free fluid; some images include measurements. Tabular clinical data include demographic characteristics, clinical history and examination findings, laboratory results, and expert ultrasonographic findings. The

reference standard was the binary diagnosis label (appendicitis vs no appendicitis) derived from histopathological examination where surgical resection was performed, and from clinical follow-up combined with scoring criteria in conservatively managed patients, as described in the original dataset publication [21].

The entire dataset contains 782 patient records. Patients without available ultrasound images, patients in whom the appendix was not visualized on ultrasound per the tabular Appendix_on_US field, patients whose image pool consisted exclusively of caliper-measurement views after removal of non-visualized images, and cases with missing or uninterpretable diagnosis labels were excluded. Patient-level assessment was used throughout to prevent information leakage across different inputs from the same examination.

Model specification

The evaluated model was Llama 4 Maverick Instruct [24] (Meta AI, Menlo Park, CA; identifier: meta-llama/Llama-4-Maverick-17B-128E-Instruct-FP8), accessed through the Together AI inference API. The multilingual, open-weight model was used in an inference-only configuration without fine-tuning, retrieval augmentation, or external tool use. Llama 4 Maverick is a mixture-of-experts multimodal language model supporting interleaved image and text-input; it was released publicly on 5 April 2025 with a reported knowledge cutoff of August 2024. The model comprises 17 billion active parameters across 128 experts; the image encoder architecture and vision-language fusion strategy are not publicly documented. The FP8-quantized checkpoint was used as accessed on and around 15 March 2026.

Inference was orchestrated from a local Python environment (together, pandas, Pillow, scikit-learn, tqdm). Preprocessing, API communication, and result evaluation were performed locally; model execution occurred on Together AI's remote infrastructure (Together AI, San Francisco, CA). Client-side execution was performed on a Windows 11 workstation with an Intel Core i5-14400F CPU (Intel Corporation, Santa Clara, CA), NVIDIA GeForce RTX 5060 8-GB GPU (NVIDIA Corporation, Santa Clara, CA), and Python 3.14.3.

Experimental design

Five experiments using different inputs, as well as combinations of inputs, were evaluated: images only, ultrasound report text only, images and ultrasound text, images and clinical data, and the entire multimodal input. Each experiment was evaluated in a single inference pass per patient to isolate the diagnostic contribution of each information channel. All experiments were applied to each patient using matched image, text report, and clinical data, ensuring full comparability. The experiments are summarized in Table 1.

Image and text preprocessing

Ultrasound images were provided in BMP format and converted to RGB. To satisfy API payload constraints, images were resized in memory (longest side $\leq$768 pixels, Lanczos resampling) and JPEG-compressed with progressive quality reduction (starting at quality 85, stepped to a minimum of 40) until the encoded size was below 650 kB per image, then base64-encoded. No annotations, masks, manual crops, or region proposals were added.

A maximum of four images per patient was submitted in experiments involving visual input. When more than four eligible images were available, a pre-specified priority rule selected: (1) appendix orientation views (longitudinal or transverse); followed by (2) secondary context views (right lower quadrant, free fluid, lymph node images). This heuristic was based on the diagnostic hierarchy of appendicitis sonography and was not empirically optimized on the study cohort.

Tabular sonographic findings (experiments 2, 3, 5) were formatted as structured natural-language text (e.g., "Appendix diameter: 8.2 mm; Free fluid: Yes; Target sign: Yes"). Clinical laboratory data (Exp 4, 5) were similarly formatted (e.g., "WBC count: 14.2×10^3/μL; CRP: 48.3 mg/L"). Only columns confirmed present in the dataset were included; missing values were recorded as "not recorded."

Table 1 Five experiments and provided inputs

Condition	Label	Input provided to the model
Exp 1	Images only	Up to four clean ultrasound images; no text or laboratory context
Exp 2	US report text only	Structured sonographic findings text (appendix diameter, free fluid, target sign, appendicolith, surrounding tissue reaction, wall layers, perfusion, bowel changes, lymph nodes); no images
Exp 3	Images+US text	Images (as Exp 1) plus structured sonographic text (as Exp 2). Text findings were designated as primary
Exp 4	Images+clinical data	Images plus demographic and laboratory variables (age, sex, body temperature, WBC count, CRP, neutrophil percentage); no sonographic text
Exp 5	Full multimodal	Images, structured sonographic text, and clinical laboratory data combined

Exp experiment, *US* ultrasound, *WBC* white blood cell, *CRP*, C-reactive protein

Prompt design

Prompt design followed a fixed rubric-based strategy that operationalized the principal interpretive steps of pediatric appendicitis ultrasound evaluation as explicit model instructions. A shared prompt governed all image-containing experiments (experiments 1, 3, 4, 5); a separate prompt was used for the text-only experiment (experiment 2). Both prompts were finalized before evaluation and were not modified thereafter. Full prompt texts are provided on the Open Science Framework [23].

For image-containing experiments, the prompt instructed the model to assess three dimensions: appendix visualization quality, direct appendiceal abnormality, and secondary signs. Four deterministic decision rules were applied in priority order. In priority order: (1) convincing direct appendix abnormality (B=2) → appendicitis; (2) equivocal appendix (B=1) with at least partial visualization (A≥1) → appendicitis, weighted toward sensitivity; (3) clearly normal appendix fully visualized (B=0, A=2) → no appendicitis; and (4) appendix partially seen without abnormality (B=0, A≤1) → no appendicitis. Confidence scores were assigned using a study-specific heuristic mapping table. Thresholds were defined to distinguish levels of diagnostic certainty according to the strength of imaging evidence rather than to represent empirically calibrated probabilities. Higher ranges were assigned when multiple direct imaging criteria were present, whereas lower ranges were used for equivocal findings (e.g., three or more criteria, 0.92–0.96; equivocal rule applied, 0.58–0.65). For the text-only experiment (experiment 2), the prompt instructed the model to apply threshold-based decision rules to the provided sonographic findings and map the number of positive primary criteria to a corresponding confidence band. These ranges were primarily designed to assign an ordinal confidence ranking to enable area under the receiver operating characteristic curve (AUROC) calculation as a discrimination metric, rather than to represent empirically calibrated probabilities.

The prompt was developed iteratively using a small set of example cases not drawn from the evaluation cohort. No metric-based optimization was performed on the study cohort, and no prompt modifications were made after evaluation began. All prompts and structured input text were composed in English.

Stochasticity control and output handling

Temperature was set to zero specifically to minimize output stochasticity and maximize run-to-run reproducibility. Maximum output token limits were set per experiment to accommodate the expected response length: 400 tokens for experiments 1 and 2, 450 tokens for experiment 4, and 700 tokens for Exp 3 and 5. A single output was generated per patient per condition. Automatic retry with exponential back-off (up to two additional attempts) was applied for API communication errors only; model outputs were not retried based on diagnostic task content. No formal safety screening of model outputs was applied.

The model was instructed to produce a brief structured reasoning statement followed by a JSON object on the final output line containing the diagnosis label and a confidence value. Labels were extracted using a parser that identified the last valid JSON object containing a diagnosis field in the response. Outputs from which no parseable diagnosis label could be extracted were marked as missing and excluded from metric calculation.

Statistical analysis

Diagnostic performance was assessed at the patient level against the reference standard. Primary analyses focused on sensitivity, specificity, accuracy, positive predictive value (PPV), negative predictive value (NPV), and F1-score. AUROC was analyzed to quantify discrimination across the confidence-score ranking. The AUROC summarizes a model's ability to rank positive cases above negative cases across all decision thresholds, ranging from 0.5 (chance-level) to 1.0 (perfect discrimination). Because confidence scores were derived from a study-specific heuristic mapping table rather than calibrated probabilities, AUROC was interpreted as a measure of relative ranking performance rather than absolute risk estimation. This interpretation is consistent with precedent in clinical scoring system literature, where AUROC is routinely derived from rubric-based scores such as the Alvarado and Pediatric Appendicitis scores rather than from model-generated probabilities [8, 25, 26]. Nonparametric bootstrap 95% confidence intervals (CIs) were estimated using 10,000 resamples with replacement. Pairwise AUROC comparisons were performed using the DeLong nonparametric test for correlated curves applied to paired predictions from the same patients. Pairwise binary prediction agreement was assessed using McNemar's test. A Bonferroni correction was applied across all ten pairwise comparisons (adjusted α=0.005). All analyses used Python. We used scikit-learn for classification metrics and bootstrap resampling; scipy.stats for McNemar's test and DeLong's test. It was implemented per DeLong et al. [27].

Results

Study population

Of 782 patients in the Regensburg Pediatric Appendicitis Dataset, 294 were retained for evaluation after sequential application of four exclusion criteria (Fig. 1). The retained cohort comprised 243 patients with confirmed appendicitis (82.7%) and 51 patients without appendicitis (17.3%). Patient demographic and imaging characteristics are summarized in Table 2. Age was similar between diagnostic groups (mean 11.2±3.4 years in the appendicitis group vs 11.0±3.5 years in

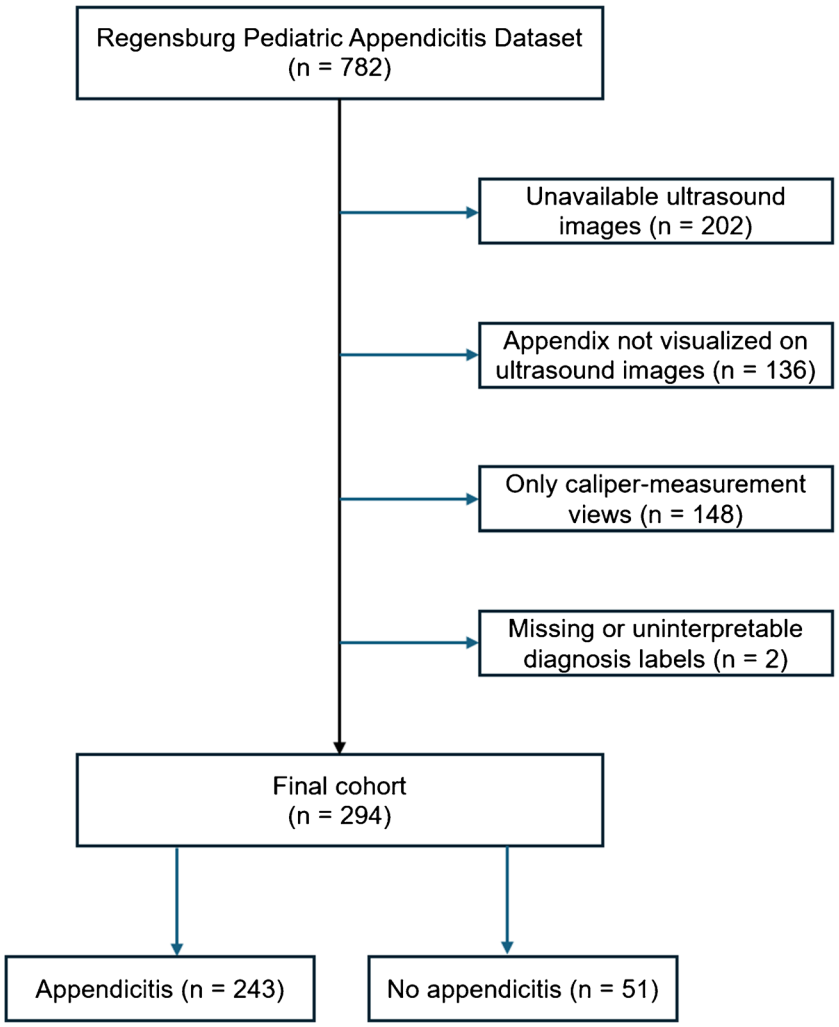


Fig. 1 Flow diagram of cohort selection from the Regensburg Pediatric Appendicitis Dataset

the no-appendicitis group). Male sex was slightly more common overall (162/294, 55.1%). The predominant age group was 5–12 years (160/294, 54.4%). The characteristic transverse ultrasound appearance of appendicitis is illustrated in Fig. 2.

Diagnostic performance across experimental conditions

One patient lacked a parsable model output in every experiment and was excluded from all evaluations (evaluable n=293 for experiments 1, 2, 4, and 5). In experiment 3 (images+US text), nine additional outputs were non-binary and were treated as missing for primary metric calculation (evaluable n=284 for Exp 3).

The lowest-performing experiment was experiment 1, which resulted in very low specificity (13.7%) despite high sensitivity (95.5%), with 44 of 51 true-negative patients misclassified as appendicitis; this condition also showed the weakest discrimination (AUROC 0.567). Providing structured sonographic findings as text, without images, increased specificity to 62.8% while maintaining sensitivity at 94.2%, with a corresponding AUROC of 0.883. Adding ultrasound images to the structured text report yielded AUROC 0.897. Adding clinical and laboratory context to images improved AUROC to 0.715, but remained substantially below text-input containing experiments (Fig. 3). The full multimodal experiment achieved the highest AUROC, at 0.902, with a sensitivity of 94.2% and a specificity of 57.7%. Complete performance metrics are presented in Table 3, and the confusion matrix is shown in Table 4.

Seven of ten DeLong comparisons reached the corrected significance threshold. All four comparisons between experiment 1 (images only) and any text-containing or laboratory-augmented experiment were significant ($P \leq 0.002$). In contrast, all three pairwise comparisons among the high-performing text-containing experiments (experiments 2, 3, 5) were non-significant (DeLong P=0.489–0.758). McNemar's test reached significance after correction only for experiment 1 vs experiment 2 (P=0.003) and experiment 1 vs experiment 3 (P=0.004). Table 5 presents the detailed results of

Table 2 Demographic and imaging characteristics of the retained cohort (*n*=294)

Characteristic	Entire cohort (*n*=294)	Appendicitis (*n*=243)	No appendicitis (*n*=51)
Age (years)			
Mean±SD	11.2 ± 3.4	11.2 ± 3.4	11.0 ± 3.5
Median (IQR)	11.3 (9.2–13.5)	11.4 (9.2–13.5)	11.0 (9.1–13.2)
Range	0.0–17.9	0.0–17.9	0.8–17.8
Sex, *n* (%)			
Male	162 (55.1%)	135 (55.6%)	27 (52.9%)
Female	131 (44.6%)	107 (44.0%)	24 (47.1%)
Unknown	1 (0.3%)	1 (0.4%)	0
Age group, *n* (%)			
0–5 years	14 (4.8%)	12 (4.9%)	2 (3.9%)
5–12 years	160 (54.4%)	130 (53.5%)	30 (58.8%)
12–18 years	120 (40.8%)	101 (41.6%)	19 (37.3%)
Ultrasound images per patient			
Total available, mean±SD	3.9 ± 2.4	4.1 ± 2.5	2.9 ± 1.6
Median (IQR)	3.5 (2.0–5.0)	4.0 (2.0–5.0)	3.0 (2.0–4.0)
Clean images available, mean±SD	2.3 ± 1.5	2.4 ± 1.5	1.8 ± 1.2
Images submitted to model, *n* patients (%)			
1 image	119 (40.5%)	90 (37.2%)	29 (56.9%)
2 images	72 (24.5%)	62 (25.5%)	10 (19.6%)
3 images	43 (14.6%)	35 (14.5%)	8 (15.7%)
4 images	60 (20.4%)	56 (23.1%)	4 (7.8%)

IQR interquartile range, *SD* standard deviation

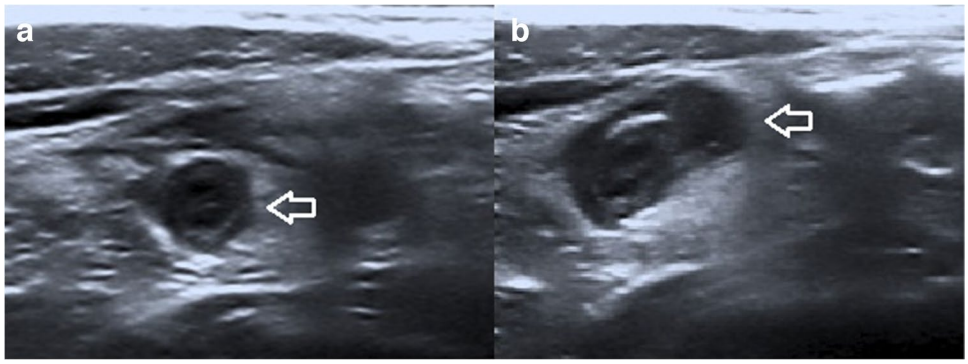


Fig. 2 Representative grayscale B-mode abdominal ultrasound images in a 5-year-old boy with appendicitis. **a** Transverse image demonstrating a rounded tubular structure with concentric wall layering (*arrow*). **b** Image from the same examination in an oblique longitudinal plane showing persistent target-like morphology (*arrow*). Licensed under CC-BY-NC 4.0

DeLong's nonparametric test for correlated AUROC curves and McNemar's test for binary prediction agreement.

Discussion

This study evaluated the zero-shot diagnostic performance of a VLM for pediatric appendicitis across five different input as well as input combinations using the publicly available Regensburg Pediatric Appendicitis Dataset. The principal finding is that raw ultrasound image interpretation by a general-purpose VLM without textual context is insufficient for clinically meaningful appendicitis diagnosis. In contrast, providing the model with structured sonographic imaging report as text yielded significantly better discrimination ($P<0.001$), and this performance was not meaningfully improved by subsequently adding images ($P=0.758$) or combining all available data sources ($P=0.489$). In other words, the model functions as an effective text-based clinical decision support system when given structured findings, but fails as a visual interpreter of pediatric ultrasound when operating on images alone.

The most important clinical implication of the image-only experiment (experiment 1) result is the specificity of 13.5%. Under this condition, 44 of 51 patients without appendicitis were incorrectly diagnosed as appendicitis, corresponding to

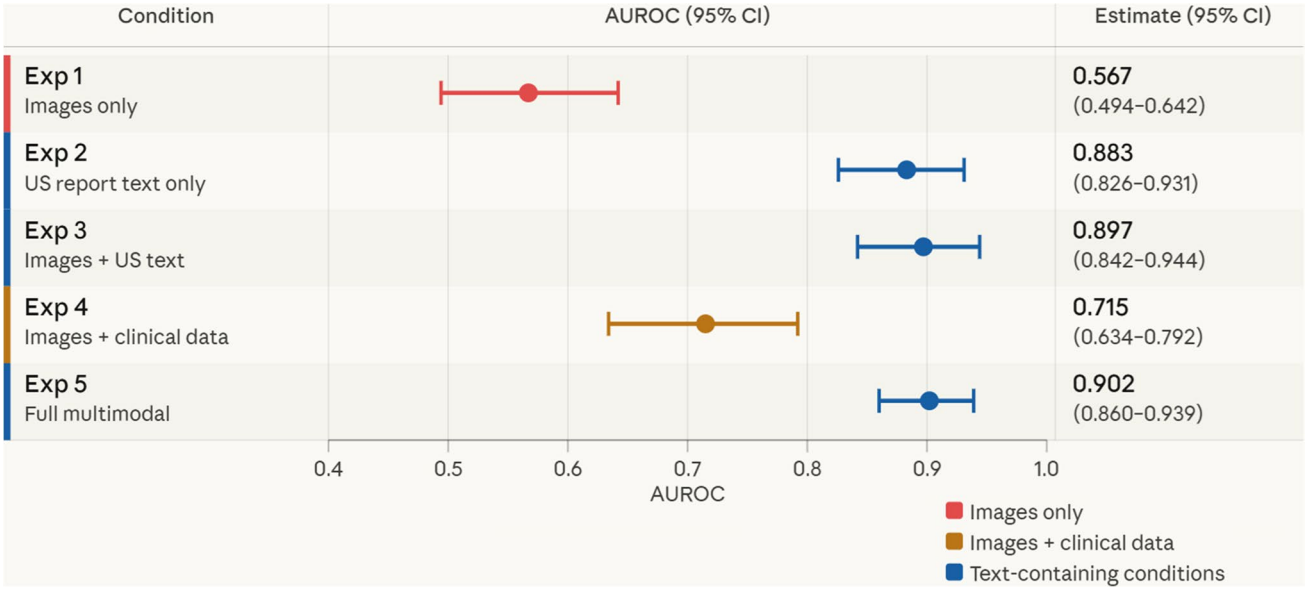


Fig. 3 Area under the receiver operating characteristic curve (AUROC) with 95% confidence intervals across five experiments

Table 3 Diagnostic performance of Llama 4 Maverick Instruct across five experiments with nonparametric bootstrap 95% confidence intervals (10,000 resamples)

Condition (*n*)	Sensitivity (95% CI)	Specificity (95% CI)	Accuracy (95% CI)	PPV (95% CI)	NPV (95% CI)	F1 (95% CI)	AUROC (95% CI)
Exp 1	95.5%	13.5%	81.2%	83.7%	38.9%	0.892	0.567
293	92.7–97.9%	4.9–23.6%	76.2–85.4%	79.3–87.9%	16.7–63.2%	0.863–0.918	0.494–0.642
Exp 2	94.2%	63.5%	88.7%	92.3%	70.2%	0.933	0.883
293	91.1–97.0%	50.0–76.8%	85.0–92.2%	88.8–95.5%	56.5–82.9%	0.909–0.954	0.826–0.931
Exp 3	94.0%	64.7%	89.1%	92.4%	70.2%	0.932	0.897
284	90.9–96.9%	51.3–77.8%	84.9–92.3%	88.9–95.7%	56.5–83.0%	0.908–0.955	0.842–0.944
Exp 4	89.7%	42.3%	81.3%	87.9%	46.8%	0.888	0.715
293	85.7–93.3%	29.1–56.1%	76.9–85.7%	83.7–91.8%	32.6–61.2%	0.857–0.916	0.634–0.792
Exp 5	94.2%	57.7%	88.1%	91.2%	68.2%	0.927	0.902
293	91.1–97.0%	43.9–71.2%	84.0–91.5%	87.6–94.6%	54.1–81.6%	0.901–0.949	0.860–0.939

AUROC area under the receiver operating characteristic curve, *NPV* negative predictive value, *PPV* positive predictive value

Table 4 Confusion matrix counts and output validity per experiment

Condition	TP	FP	FN	TN	Missing[a]	Abstentions[b]
Exp 1 (*n*=293)	231	44	11	7	1	0
Exp 2 (*n*=293)	228	19	14	32	1	0
Exp 3 (*n*=284)	220	17	14	33	1	9
Exp 4 (*n*=293)	217	29	25	22	1	0
Exp 5 (*n*=293)	228	21	14	30	1	0

FN false negative, *FP* false positive, *TN* true negative, *TP* true positive

[a]Outputs from which no parseable binary label could be extracted, excluded from all metrics

[b]Non-binary outputs (e.g., "indeterminate," "equivocal") occurring exclusively in Exp 3

Table 5 Pairwise area under the receiver operating characteristic curve (*AUROC*) differences and statistical tests across all 10 condition pairs

Condition A	Condition B	ΔAUROC	DeLong *P*	McNemar *P*
Exp 1	Exp 2	0.316	**<0.001**	**0.003**
Exp 1	Exp 3	0.323	**<0.001**	**0.004**
Exp 1	Exp 4	0.149	**0.002**	1.000
Exp 1	Exp 5	0.335	**<0.001**	**0.007**
Exp 2	Exp 3	0.008	0.758	0.831
Exp 2	Exp 4	−0.168	**0.002**	**0.010**
Exp 2	Exp 5	0.019	0.489	0.677
Exp 3	Exp 4	−0.182	**<0.001**	**0.008**
Exp 3	Exp 5	0.013	0.542	1.000
Exp 4	Exp 5	0.186	**<0.001**	0.017

DeLong nonparametric test for correlated ROC curves; McNemar's test with Edwards continuity correction for binary prediction agreement. Bonferroni-adjusted significance threshold α=0.005 (10 pairs)

a false-positive rate of 86.5%. In a real deployment context, this would translate to a large excess of unnecessary surgical consultations, computed tomography examinations, or operative interventions in children correctly managed non-operatively. This result confirms that zero-shot visual deployment of general-purpose VLMs for autonomous ultrasound diagnosis of pediatric appendicitis is unreliable in its current form and without structured textual grounding. The strong performance of the text-only condition (experiment 2) reframes the appropriate role of VLMs in this clinical domain. Rather than replacing or augmenting the radiologist's visual interpretation, a VLM such as Llama 4 Maverick is better suited to consuming the radiologist's structured text report and providing decision support for the clinician who must integrate sonographic text findings with clinical context. A secondary implication concerns the role of laboratory data. Experiment 4 demonstrated that adding age, sex, body temperature, WBC count, CRP, and neutrophil percentage to image input improved AUROC from 0.567 to 0.715 (P=0.002), but this remained far below the text-containing experiments (P=0.002). That suggests that the anatomical specificity of sonographic findings is irreplaceable by systemic inflammatory markers, even when a VLM integrates them.

The most direct benchmark for this study is the work of Marcinkevičs et al. [21], who developed and evaluated a model on the same Regensburg Pediatric Appendicitis Dataset using clinical tabular features and reported an AUROC of approximately 0.80 for appendicitis diagnosis. Our text-only experiment (experiment 2, AUROC 0.883) and full multimodal experiment (experiment 5, AUROC 0.902) numerically exceed this benchmark, though direct comparison must be made cautiously. Marcinkevičs et al.'s study used the full dataset including patients excluded here, employed a supervised trained model rather than zero-shot inference, and had access to a richer set of clinical variables. The exclusion of 148 patients with only caliper-measurement images from our cohort likely enriched our retained cohort for cases with cleaner diagnostic profiles, which may have inflated our observed performance.

The strong text-only experiment performance (AUROC 0.883) is consistent with the broader finding that large language models are highly capable clinical decision support tools when provided with structured clinical data. Studies evaluating GPT and Gemini models on structured clinical vignettes for appendicitis triage have demonstrated that strong baseline awareness of contemporary acute appendicitis management strategies [28]. Our experiment 2 result extends this literature by demonstrating that structured sonographic findings alone without clinical scores, demographics, or laboratory values provide sufficient information for near-guideline-level discrimination. This suggests that structured reporting of ultrasound findings may be a sufficient interface between the radiologist and an LLM-based decision support layer.

Several limitations of this study must be acknowledged. First, the prompt design involved iterative informal development on a small set of cases outside the evaluation cohort. While no metric-based optimization was performed on the study cohort and prompts were not modified after evaluation began, the subjective choices embedded in the rubric, the dimensional scoring system, decision rule priority order, and confidence table thresholds represent a form of prompt engineering that cannot be fully disentangled from model capability. The extent to which these choices drove performance, particularly in the text-only experiment where the prompt's criteria-counting approach closely mirrors threshold-based clinical scoring, cannot be formally quantified without a separately held-out prompt development set. In the text-only experiment in particular, the prompt's step-by-step criteria-counting logic closely mirrors a clinical scoring system; the model's role may therefore be closer to executing pre-specified rules than to independent clinical reasoning. Second, the exclusion of 148 patients whose image pools consisted entirely of caliper-measurement views introduced potential selection bias. Sonographers typically acquire measurement images when they identify an abnormal structure, meaning excluded patients are likely enriched for true appendicitis. Conversely, the retained cohort may be enriched for cases where the appendix was clearly visualized and the sonographic findings were unambiguous, which could have favored performance in both the image-based and text-based experiments. Third, the appendicitis prevalence of 82.7% in the retained cohort substantially exceeds the estimated 15–30% prevalence among children presenting to the emergency department with acute abdominal pain [2]. The positive and negative predictive values reported in this study are therefore not generalizable to real-world clinical

deployment without prevalence adjustment. Fourth, confidence scores used to compute AUROC were derived from a protocol-embedded rubric rather than from raw model logits or calibrated probability outputs, consistent with the study's designation of AUROC as a ranking metric interpreted analogously to rubric-derived clinical scoring system AUROCs [8, 25, 26]. Accordingly, AUROC should be interpreted as a ranking-based measure of relative discrimination across conditions rather than as evidence of calibrated risk estimation or direct clinical utility. Fifth, this study was conducted on a single retrospective dataset from one European children's hospital collected over a specific time period using two ultrasound systems. Generalizability to other patient populations, healthcare settings, ultrasound equipment, image acquisition protocols, and sonographer practices is unknown. Furthermore, as the dataset was published in February 2023, before the reported knowledge cutoff of Llama 4 Maverick (August 2024), a theoretical risk of dataset contamination cannot be excluded; however, given the benchmark and tabular nature of the dataset and the zero-shot inference setting used here, unintended memorization of case-level labels is considered unlikely. External validation on independent, prospectively collected datasets is required before any clinical deployment. Sixth, the model was accessed through a commercial inference API at a specific point in time. API-level model versioning is not user-controllable, and the specific checkpoint used may be updated, deprecated, or replaced without notice. Result reproducibility across API updates cannot be guaranteed, and the specific model version should be considered a fixed experimental condition rather than a durable system. All data submitted to the API comprised publicly available, fully anonymized records with no patient identifiers transmitted. No interaction logging or formal governance framework beyond the IRB exemption was in place.

Conclusion

This modality ablation study demonstrates that zero-shot visual interpretation of pediatric abdominal ultrasound by a general-purpose multimodal large language model is insufficient for reliable appendicitis diagnosis when images are provided without textual context. When provided with structured sonographic text findings, the same model achieves substantially better performance, which is not further improved by adding raw images or laboratory data. These findings argue against deploying general-purpose VLMs as autonomous visual interpreters of pediatric ultrasound and support their use as text-based decision support tools that consume structured radiological reports. Future work should focus on fine-tuning or retrieval augmentation strategies that could close the gap between zero-shot visual performance and text-based performance.

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Data availability The Regensburg Pediatric Appendicitis Dataset (version 1.0), used in this study, is publicly available as cited in Reference 21. Additional derived analysis files are available from the corresponding author on reasonable request.

Declarations

Conflicts of interest The authors declare no competing interests.

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